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Roll No.

TEC-659

**B. TECH. (ECE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

ADVANCED EMBEDDED SYSTEM

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the different applications of Embedded Systems and General Computing Systems. (CO1)
- (b) What are Quality Attributes of Embedded System ? (CO1)
- (c) An embedded system-based visitor counter is needed. The system must be designed such that it can count only upto 100 visitors and after than it must blink the RED led. Explain the Embedded System design process for such system with the help of proper block diagram, flow chart for algorithm etc. (CO1)

P. T. O.

2. (a) What are addressing modes in 8051 microcontrollers ? Explain with the help of examples. (CO2)
- (b) Explain the PIC microcontroller with the help of its features and architectures. (CO2)
- (c) Using Interrupt programming create a delay of 500 ms using timers and blink LED connected to P2.0 with the same delay. Draw a proper schematic and write code for the given problem. (CO2)
3. (a) Explain Serial Communication of 8051. What are the pins used for serial communication of 8051 and what are the registers need to configure ? Explain also that how to adjust the baud rate of the 8051. (CO3)
- (b) A LED 7-segment display is connected to P0. Write a program to count from 0 to 9 with specific interval of 1 sec. (CO6)
- (c) Explain programming framework in Embedded System Design. (CO3)
4. (a) What is the role of Real Time Operating Systems (RTOS) in Embedded System Design ? (CO4)
- (b) What are Threads ? What are their role in Embedded System ? (CO4)
- (c) Explain process and task in RTOS. (CO4)

(3)

5. (a) Draw the architecture of MC68HC11/12 and explain its blocks in detail. (CO5)
- (b) What is ARM 32 microprocessor ? Explain in detail with the help of a diagram. (CO5)
- (c) What are AVR family microcontrollers ? Explain AT Tiny microcontroller with the help of its architecture. (CO6)

Answer

10/06/2022

Roll No.

TOE-610

**B. TECH. (ECE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

OOPs USING C++

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Write a C++ program to overload unary ++ operator without using friend function. (CO1, CO2, CO3, CO4, CO5)

(b) Write a C++ program to create a class called Test with the following members : (CO1, CO2, CO3, CO4, CO5)

Data Members :

string str

Function Members :

Void inputVal(string) → member function will store value into str.

Void showVal() → This member function will print value of str.

P. T. O.

Void checkVal() → This member function will check whether first and second last character of str string is equal or not.

In main function create object of class Test and perform the above task.

- (c) Explain Static Binding and Dynamic Binding with the help of C++ program. (CO1, CO2, CO3, CO4, CO5)
2. (a) Explain different forms of inheritance. Can private members of the base class be accessed by the derived class ? If yes, how ? (CO3, CO4, CO6)
- (b) Explain exception in C++. How it is handled in C++ ? Describe all the keywords involved in exception handling by implementing it with a C++ program. (CO3, CO4, CO6)
- (c) Write a C++ program to overload binary + operator. (CO3, CO4, CO6)
3. (a) Implement by creating a class called 'Time' that has three integer data members for hours, minutes and seconds. Define a member function to read the values, member function to add time, member function to display time in HH:MM:SS format. Write a C++ program to create two time objects, add them and display the results in HH:MM:SS format. (CO1, CO2, CO4)
- (b) What is multiple inheritance ? When does ambiguity arise in multiple inheritance ? How one resolves it ? (CO1, CO2, CO4)
- (c) Explain friend function. Develop a C++ program to swap the value of data members present in two different classes with the help of friend function. (CO1, CO2, CO4)

4. (a) (i) What is "this" pointer ? Give an example to illustrate the use of it in C++.

(ii) Differentiate between static member and instance member.

(CO1, CO2, CO4, CO5)

(b) Define a class named Test Data with the following description :

(CO1, CO2, CO4, CO5)

Data Members

private int num1 ,num2'

Constructor TestData(int, int) Constructor will store value in num1, num2.

member function void buzzno() → This member function will find buzz number between num1 and num2

Note : Condition for buzz number are (i) Divisible by 7 or (ii) Last digit is 7 or (iii) Sum of digit is 7.

(c) Create an abstract class ValidateData with the following pure virtual function :

(CO1, CO2, CO4, CO5)

virtual void checkdata(string)=0;

Create a class Person which inherits abstract class ValidateData and override the function.

void checkdata(String) method will check whether all alphabets are in uppercase or not.

5. (a) What are the differences between virtual function and pure virtual function in C++ ? Explain with the help of C++ program. (CO4, CO5)
- (b) Write a C++ program to read the content from a file and again write the same content in another file. Assume the name of source file is myfile.txt and destination file name is newfile.txt. (CO4, CO5)
- (c) Describe Diamond Problem in Inheritance. Provide the solution to the diamond problem with a C++ code. (CO4, CO5)

Roll No.

TEC-604

B. TECH. (ECE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022
DATA COMMUNICATION NETWORKS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Draw neat and clean diagram of OSI model and explain the function of each layer. (CO1)
- (b) Differentiate between an analog and a digital electromagnetic signal. Which of the three analog-to-analog conversion techniques is the most susceptible to noise ? Defend your answer. (CO1)
- (c) A multiplexer combines four 100 kbps channels using a time slot of 2 bits. Show the output with four arbitrary inputs. What is the frame rate ? What is the frame duration and What is the bit rate ? What is the bit duration ? (CO1)

P. T. O.

2. (a) Compare and contrast HDLC with PPP. Which one is byte-oriented; which one is bit-oriented? (CO2)
- (b) Compare and contrast the Go-Back-NARQ Protocol with Selective-Repeat ARQ. (CO2)
- (c) (i) Find the minimum Hamming distance of the coding scheme in Table :

Datawords	Codewords
00	000
01	011
10	101
11	110

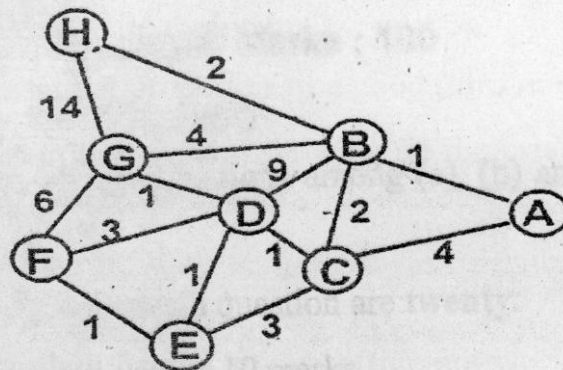
- (ii) A code scheme has a Hamming distance $d_{\min} = 4$. What is the error detection and correction capability of this scheme? (CO2)
3. (a) (i) Compare the frame structures for 10Base T, 100Base T and Giga bit Ethernet. How do they differ?
- (ii) In analyzing ALOHA, we use only one parameter, time; in a analyzing CSMA, we use two parameters, space and time. Explain the reason, why? (CO3)
- (b) Differentiate between pure ALOHA and slotted ALOHA with the help of a diagram. (CO3)
- (c) Write short notes on the following : (CO3)
- (i) IEEE Standard 802.4

(3)

(ii) IEEE Standard 802.5

(iii) IEEE Standard 802.6

4. (a) Discuss the various channel allocation schemes in detail. (CO4)
- (b) What do you understand by congestion ? Explain its types. (CO4)
- (c) Consider the network shown in Fig. below, with the indicated link costs. Use Dijkstra's shortest path algorithm to compute the shortest path from F to all network nodes. Show how the algorithm works by computing a table : (CO4)



5. (a) Discuss the term data compression. Differentiate between various data compression techniques. (CO5, CO6)
- (b) Explain Multiprotocol Label Switching (MPLS). (CO5, CO6)
- (c) Why do we need a DNS system, when we can directly use an IP address ? If a DNS domain name is *voyage.fhda.edu*, how many labels are involved here ? How many levels of hierarchy are there ? (CO5, CO6)

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TEC-603

**B. TECH. (ECE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

VLSI TECHNOLOGY AND DESIGN**Time : Three Hours****Maximum Marks : 100**

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain oxidation importance and types of oxidations in VLSI ? (CO1)
- (b) What do you understand by electronic grade silicon, and CZ crystal growth technique ? (CO1)
- (c) Explain in brief about Deal and Grove's model. (CO1)
2. (a) Explain Physical vapor Deposition and E-beam Evaporation techniques. (CO2)
- (b) Explain electroplating technique and metallization applications in VLSI. (CO2)
- (c) Explain in brief about Annealing and their applications in VLSI. (CO2)

P. T. O.

(2)

3. (a) Consider an n-channel metal oxide semiconductor field effect transistor (MOSFET) with gate to source voltage of 1.8 V. Assume that $W/L = 4$, $\mu_n C_{ox} = 70 \times 10^{-6} \text{ AV}^{-2}$, and the threshold voltage is 0.3 V. In the saturation region, find the drain conduction g_d (in micro Siemens). (CO3)
- (b) Derive the expression for current vs. voltage for MOS in non-saturation and saturation mode. (CO3)
- (c) The slope of the I_D vs. V_{GS} curve of an n-channel MOSFET in linear region is $10^{-3} \Omega^{-1}$ at $V_{DS} = 0.1 \text{ V}$. For the same device, neglecting channel length modulation, the slope of the $\sqrt{I_D}$ vs. V_{GS} curve (in $\sqrt{\text{AN}}$) under saturation region is approximately. (CO3)
4. (a) Explain Resistive load inverter. (CO4)
- (b) Explain briefly CMOS inverters. (CO4)
- (c) Explain static characteristics of MOS inverter. (CO4)
5. (a) Draw the stick diagram of expression $\{(BD) + (AC) + (EF)\}$. (CO5, CO6)
- (b) Draw step by step layout for complement of $\{(A + B)C\}$. (CO5, CO6)
- (c) What are the various industrial applications of CMOS ? Explain in brief. (CO5, CO6)

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TEC-602

**B. TECH. (EC) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

MICROWAVE ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Write short notes on cut-off, evanescent and propagation modes of waveguide. (CO1)

(b) Derive field equations for TE mode in rectangular waveguide. (CO1)

(c) A wave is propagating in a rectangular waveguide operating in TE_{10} mode at a frequency of 6 GHz. The width of the waveguide is 5 cm.

Find the cut-off wave length and wavelength within guide. (CO1)

P. T. O.

2. (a) Derive the expressions for S matrix. What is insertion loss and reflection loss ? (CO2)
- (b) Explain the propagation through an isolator with the help of a diagram. (CO2)
- (c) A 20 MW signal is fed into one of the collinear port (port 1) of a lossless H-plane T-junction. Calculate the power delivered through each port when other ports are terminated in matched load. (CO2)
3. (a) Explain the working of Gunn diode with the help of two valley models and show the negative resistance in their V-I characteristics. (CO3)
- (b) Explain the tunneling effect with the proper energy band diagrams. Write the various advantages of Tunnel Diode. (CO3)
- (c) A reflex klystron is operated at 9 GHz with a DC beam voltage of 600 V for $1\frac{3}{4}$ mode, repeller space length of 1 mm and DC beam current of 10 mA. The beam-coupling coefficient is assumed to be 1. Calculate the repeller voltage, electronic efficiency and output power. (CO3)
4. (a) Explain the principle of the power meter single bridge circuit. Why double bridge circuit is better than single bridge circuit ? (CO4)
- (b) Explain the network analyzer with help of a block diagram. (CO4)
- (c) The signal power at the input of a device is 10 MW. The power at the output of the same device is 0.20 MW. Calculate the insertion loss in dB of this component. (CO4)

(3)

5. (a) Explain the Richards transformation and Kuroda's identify.

(CO5, CO6)

- (b) Explain the effect of microwaves on human body.

(CO5, CO6)

- (c) Design a stepped impedance low pass filter with $f_c = 2$ GHz and $Z_0 = 50 \Omega$, $Z_1 = 10 \Omega$, $Z_n = 150 \Omega$, order 5.

$$g_1 = g_5 = 0.618; g_2 = g_4 = 1.618; g_3 = 2.0; g_6 = 1.$$

(CO5, CO6)

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TEC-601

B. TECH. (ECE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022
WIRELESS COMMUNICATION

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.

1. (a) A cellular system experiences path loss exponent to be 4, where a UE requires minimum 18 dB SIR to have acceptable voice quality. If there 7 cells in a cluster, will it support the minimum required SIR ? Now, assume the minimum required SIR is increased to 20 dB. Find the appropriate cluster size. (CO1)
- (b) (i) Calculate the area of a hexagonal cell of radius R.
(ii) Explain meaning of the terms 'universal frequency reuse' and 'fractional frequency reuse'. (CO1)
- (c) How is cell splitting performed ? Why can it increase the system capacity ? (CO1)

P. T. O.

2. (a) In what ways the expression obtained for received signal power in two-ray ground reflection model differ from free space propagation model ?
(CO2)
- (b) (i) What do you mean by hand-off in a cellular system ? Use power level consideration to explain the concept.
- (ii) The service area in a cellular system requires 12 repetitions of a cluster having 7 cells. Each cell has 16 channels. Find the total number of channels that can be simultaneously accessed in the system.
(CO2)
- (c) Consider a transmitter operating at 1800 MHz transmits 4 watt power. Assume path loss exponent to be 4, shadow effect of 10.5 dB and the power at reference point ($d_0 = 100$ m) is -32 dB. Find the received power at a distance of 3 km from transmitter and the allowable path loss.
(CO2)
3. (a) What do you mean by power delay profile ? Sketch typical power delay profile to explain different delay parameters of wireless mobile environment.
(CO3)
- (b) Explain different types of small-scale fading using suitable diagrams.
(CO3)
- (c) Assume mobile equipment operates over 900 MHz frequency and moves with a velocity of 96 kmph. Find the data rate to receive distortionless transmission.
(CO3)
4. (a) What do you mean by Coherence time and Doppler spread ? Use suitable mathematical derivations to explain.
(CO4)

(3)

- (b) What do you mean by Rayleigh fading model ? Derive an expression to represent PDF of SNR under Rayleigh fading model. (CO4)
- (c) How does Maximal Ratio Combining function ? Derive an expression to represent mean SNR improvement realized in Maximal Ratio Combining. (CO4)
5. (a) State and explain the properties of maximum length sequence. (CO5 and CO6)
- (b) Draw the block diagram of direct sequence spread spectrum (DSSS) system transmitter and receiver and explain the function of each block. (CO5 and CO6)
- (c) Draw the block diagram of Wireless Communication System and explain the function of each block. (CO5 and CO6)